Item	Content
Document ID	OAM-L-BENCH-RESULT-OAML2026102301-R01
Report language	English
Serial / arm side	OAML2026102301 / Left arm L
Generated at	2026-10-26T11:07:28+08:00
Verdict	**PASS**
Operator	xlc
Project Lead	pengcheng
Workstation version	0.10.0
Criteria version	2026-10-21.3
LeRobot	0.6.2 (https://github.com/huggingface/lerobot@eb7a54c6ff49)
Hardware references	pkooij/open-arms-mini README/BOM; lerobot teleoperators/openarm_mini
Power supply	7.5 V (official BOM), tested after full-arm assembly

Test object: fully assembled OpenArm Mini Leader arm. Motor IDs were assigned individually after assembly; motor torque remained disabled throughout testing.

1 Release verdict

Stage	Test item	Result	Recorded at	Notes
S1	Preparation checklist	**PASS**	2026-10-23T11:21:50+08:00	
S2	J8 • Gripper write ID	**PASS**	2026-10-23T10:34:37+08:00	
S2	J8 • Gripper rescan after power cycle	**PASS**	2026-10-23T10:34:46+08:00	
S2	J8 • Gripper confirmed by moving the joint	**PASS**	2026-10-23T10:35:08+08:00	span 2088 counts
S2	J7 • Wrist Roll write ID	**PASS**	2026-10-23T10:36:16+08:00	
S2	J7 • Wrist Roll rescan after power cycle	**PASS**	2026-10-23T10:36:23+08:00	
S2	J7 • Wrist Roll confirmed by moving the joint	**PASS**	2026-10-23T10:36:32+08:00	span 2017 counts
S2	J6 • Wrist Flex write ID	**PASS**	2026-10-23T10:37:16+08:00	
S2	J6 • Wrist Flex rescan after power cycle	**PASS**	2026-10-23T10:37:26+08:00	
S2	J6 • Wrist Flex confirmed by moving the joint	**PASS**	2026-10-23T10:37:35+08:00	span 1855 counts
S2	J5 • Forearm Rotation write ID	**PASS**	2026-10-23T10:38:10+08:00	
S2	J5 • Forearm Rotation rescan after power cycle	**PASS**	2026-10-23T10:38:18+08:00	
S2	J5 • Forearm Rotation confirmed by moving the joint	**PASS**	2026-10-23T10:38:27+08:00	span 2046 counts

Stage	Test item	Result	Recorded at	Notes
S2	J4 • Elbow Flex write ID	PASS	2026-10-23T10:38:46+08:00	
S2	J4 • Elbow Flex rescan after power cycle	PASS	2026-10-23T10:38:53+08:00	
S2	J4 • Elbow Flex confirmed by moving the joint	PASS	2026-10-23T10:39:13+08:00	span 2017 counts
S2	J3 • Shoulder Roll write ID	PASS	2026-10-23T10:39:30+08:00	
S2	J3 • Shoulder Roll rescan after power cycle	PASS	2026-10-23T10:39:36+08:00	
S2	J3 • Shoulder Roll confirmed by moving the joint	PASS	2026-10-23T10:39:45+08:00	span 2887 counts
S2	J2 • Shoulder Lift write ID	PASS	2026-10-23T10:39:58+08:00	
S2	J2 • Shoulder Lift rescan after power cycle	PASS	2026-10-23T10:40:05+08:00	
S2	J2 • Shoulder Lift confirmed by moving the joint	PASS	2026-10-23T10:40:13+08:00	span 2220 counts
S2	J1 • Shoulder Pan write ID	PASS	2026-10-23T10:40:31+08:00	
S2	J1 • Shoulder Pan rescan after power cycle	PASS	2026-10-23T10:40:33+08:00	
S2	J1 • Shoulder Pan confirmed by moving the joint	PASS	2026-10-23T10:40:48+08:00	span 1577 counts
S3	Eight-axis electrical check	PASS	2026-10-23T10:51:40+08:00	8/8 passed
S3	Communication stability	PASS	2026-10-23T11:23:13+08:00	11631 communication polling cycles, 0 failures
S4	J1 • Shoulder Pan joint identity	PASS	2026-10-23T10:54:22+08:00	
S4	J2 • Shoulder Lift joint identity	PASS	2026-10-23T11:12:01+08:00	
S4	J3 • Shoulder Roll joint identity	PASS	2026-10-23T11:12:13+08:00	
S4	J4 • Elbow Flex joint identity	PASS	2026-10-23T11:12:37+08:00	
S4	J5 • Forearm Rotation joint identity	PASS	2026-10-23T11:12:49+08:00	
S4	J6 • Wrist Flex joint identity	PASS	2026-10-23T11:13:01+08:00	
S4	J7 • Wrist Roll joint identity	PASS	2026-10-23T11:13:18+08:00	
S4	J8 • Gripper joint identity	PASS	2026-10-23T11:13:32+08:00	
S5	Calibration precheck	PASS	2026-10-23T15:24:44+08:00	
S5	Official calibration saved	PASS	2026-10-23T15:25:57+08:00	
S5	Consistency check	PASS	2026-10-23T15:26:03+08:00	

Stage	Test item	Result	Recorded at	Notes
S5	Consistency check after power cycle	PASS	2026-10-23T15:26:42+08:00	
S6	Official readout: zero pose	PASS	2026-10-23T15:26:49+08:00	
S6	Official readout: gripper open	PASS	2026-10-23T15:29:13+08:00	
S6	Official readout: J6 · Wrist Flex sweep	PASS	2026-10-23T15:29:54+08:00	
S6	Official readout: J7 · Wrist Roll sweep	PASS	2026-10-23T15:30:26+08:00	

2 S1 Preparation

Checklist item	Result
Supply 7.50 V / ⩾2 A (official BOM), red positive, black negative, OCP off, OUTPUT on	✔
The arm is fully assembled and mounted on a vertical post so it hangs freely (as in the official photo)	✔
Waveshare driver board connected to the 7.5 V supply and by USB to this computer	✔
Waveshare driver board jumper set to B	✔
Gripper parts (J8 L/R, trigger L/R, J8 holder L/R) match the arm side in the serial	✔
Recorded at	2026-10-23T11:21:50+08:00

3 S2 Write each ID after assembly

Motor IDs were assigned individually after assembly and verified by a power-cycle rescan and joint-position feedback.

Joint	Original ID	Written ID	Power-cycle rescan	Confirmation span	Voltage V	Firmware	Confirmation
J8 · Gripper	1	8	Yes	2088	7.3	3.10	Passed
J7 · Wrist Roll	1	7	Yes	2017	7.3	3.10	Passed
J6 · Wrist Flex	1	6	Yes	1855	7.3	3.10	Passed
J5 · Forearm Rotation	1	5	Yes	2046	7.4	3.10	Passed
J4 · Elbow Flex	1	4	Yes	2017	7.3	3.10	Passed
J3 · Shoulder Roll	1	3	Yes	2887	7.2	3.10	Passed
J2 · Shoulder Lift	1	2	Yes	2220	7.3	3.10	Passed
J1 · Shoulder Pan	1	1	Yes	1577	7.4	3.10	Passed

4 S3 Full-arm electrical and communication tests

Joint	ID	Position	Voltage V	Temperature °C	Status	Torque	Firmware	Over-voltage limit V	Over-temperature limit °C	Phase	Max_Acc	Judgement
J1 · Shoulder Pan	1	831	7.3	28	0	0	3.10	12	80	92	250	Passed
J2 · Shoulder Lift	2	2881	7.3	27	0	0	3.10	12	80	92	250	Passed
J3 · Shoulder Roll	3	2298	7.3	28	0	0	3.10	12	80	92	250	Passed

Joint	ID	Position	Voltage V	Temperature °C	Status	Torque	Firmware	Over-voltage limit V	Over-temperature limit °C	Phase	Max_Acc	Judgement
J4 · Elbow Flex	4	761	7.2	29	0	0	3.10	12	80	92	250	Passed
J5 · Forearm Rotation	5	2673	7.3	28	0	0	3.10	12	80	92	250	Passed
J6 · Wrist Flex	6	1257	7.2	28	0	0	3.10	12	80	92	250	Passed
J7 · Wrist Roll	7	2120	7.3	29	0	0	3.10	12	80	92	250	Passed
J8 · Gripper	8	2777	7.2	28	0	0	3.10	12	80	92	250	Passed

Supply profile mini_field_7v5; firmware 3.10; voltage spread 0.1 V (recorded only). Recorded at 2026-10-23T10:51:40+08:00.

Communication stability: 60 s of continuous polling; 11631 communication polling cycles, 93048 motor-register reads, 0 failures. Polling cycles are not mechanical movements or separate tests.

5 S4 Joint identity and mobility

Tested joint	Span (counts)	Motor with largest span	Runner-up span	Judgement
J1 · Shoulder Pan	972	J1 · Shoulder Pan	134	Passed
J2 · Shoulder Lift	1811	J2 · Shoulder Lift	86	Passed
J3 · Shoulder Roll	3151	J3 · Shoulder Roll	106	Passed
J4 · Elbow Flex	1732	J4 · Elbow Flex	108	Passed
J5 · Forearm Rotation	2572	J5 · Forearm Rotation	280	Passed
J6 · Wrist Flex	904	J6 · Wrist Flex	249	Passed
J7 · Wrist Roll	2023	J7 · Wrist Roll	120	Passed
J8 · Gripper	2414	J8 · Gripper	51	Passed

Criterion: the motor with the largest span must be the tested joint and ⩾ runner-up × 3; tested span ⩾ 100 counts.

6 S5 Official calibration and consistency check

Joint	id	drive_mode	homing_offset	range_min	range_max
J1 · Shoulder Pan	1	0	-1206	0	4095
J2 · Shoulder Lift	2	0	838	0	4095
J3 · Shoulder Roll	3	0	-760	0	4095
J4 · Elbow Flex	4	0	-1254	0	4095
J5 · Forearm Rotation	5	0	1279	0	4095
J6 · Wrist Flex	6	0	-717	0	4095
J7 · Wrist Roll	7	0	-34	0	4095
J8 · Gripper	8	0	1637	2047	3732

Calibration saved at 2026-10-23T15:25:57+08:00.

7 S6 Official-driver readout verification

Time	Mode	Joint	Readings	Judgement
2026-10-23T15:26:49+08:00	Zero pose		S6.calibrated=True; S6.zero_pose=2.86; S6.gripper_closed=0	Passed
2026-10-23T15:29:13+08:00	Gripper open		S6.calibrated=True; S6.gripper_open=-65	Passed
2026-10-23T15:29:54+08:00	Sweep	J6 · Wrist Flex	S6.calibrated=True; S6.remap=145.93; S6.direction=+	Passed
2026-10-23T15:30:26+08:00	Sweep	J7 · Wrist Roll	S6.calibrated=True; S6.remap=177.23; S6.direction=-	Passed

Customer teleoperation command: do not copy and run it unchanged. First activate the LeRobot environment, replace <can0> with the Follower CAN interface and <follower_id> with the Follower ID, and verify that --teleop.port matches the Mini serial port on the customer's computer. The command can then be copied and run.

```
lerobot-teleoperate \
    --robot.type=openarm_follower --robot.port=<can0> --robot.side=left --robot.id=<follower_id> \
    --teleop.type=openarm_mini --teleop.port=/dev/serial/by-id/usb-1a86_USB_Single_Serial_5959112505-if00 \
    --teleop.side=left --teleop.id=OAML2026102301
```

8 Release criteria

ID	Description	Limit	Basis
S1.checklist	Preparation checklist fully confirmed, operator recorded	all	Official
S2.single_on_bus	Broadcast ping answered by exactly one motor, model 777	1	Official
S2.voltage	Measured voltage within the tolerance of 7.5 V and below rated + tolerance	1 V	Derived
S2.persisted	Found again at the new ID and 1 Mbps after ⩾5 s power off	rescan	Derived
S2.joint_confirmed	Moving only this joint changes the reading by ⩾ threshold (the ID sits on this joint's motor)	100 counts	Derived
S3.present	IDs 1–8 all answer, model 777	8	Official
S3.firmware_same	Same firmware on all 8 (checked by LeRobot on connect)	same	Official
S3.voltage	Each voltage within the tolerance of 7.5 V	1 V	Derived
S3.temperature	Temperature at rest, torque off ⩽ threshold	40 °C	Derived
S3.status	Status register is 0 (no alarm bits)	0	Derived
S3.torque_off	Torque disabled	0	Derived
S3.comm	Continuous reads of all 8 positions, 0 failures	60 s	Derived
S4.identity	When one joint is moved its travel is largest and ⩾ runner-up × threshold	3	Derived
S4.span	Tested joint travel ⩾ threshold	100 counts	Derived
S5.precheck	Precheck: single-turn readings, predicted offset | · | ⩽ 2047	2047	Official
S5.saved	Official calibration printed the save marker	saved	Official
S5.consistent	Calibration file and EEPROM consistent 8/8, Phase=12	8	Official
S5.consistent_power_cycled	Consistent 8/8 again after ⩾5 s power off	8	Derived
S6.calibrated	Official driver reports calibrated (is_calibrated)	is_calibrated	Official
S6.zero_pose	At zero pose J1–J7 readings | · | ⩽ threshold	15 °	Derived
S6.gripper_closed	Gripper closed reading ⩾ threshold (near 0°)	-10 °	Official
S6.gripper_open	Gripper open reading ⩽ threshold (near −65°)	-50 °	Official
S6.remap	Sweeping J6/J7 changes the swapped key most	joint_6<->joint_7	Official

9 Delivery note

This report covers factory tests S1–S7. Follower pairing and other project-specific tests are performed separately when required for delivery.

10 Sign-off

Role	Name	Signature	Date
Tester	xlc		
Project Lead	pengcheng		
Reviewer			